

INTERACTIVE QUIZ APP

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COURSE: UI/UX Design Fundamentals

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DATE OF SUBMISSION: 27-09-2025

Abstract

Defining the structure of the document, presenting a fully responsive, single-page web application that provides an interactive quiz experience. Users can select quiz topics, answer multiple-choice questions, receive instant feedback, track their scores, and view a summary at the end. The app also supports timed quizzes and progress tracking. Built using HTML5, CSS3, Bootstrap, and JavaScript (jQuery), the site delivers a modern and engaging user experience while maintaining high responsiveness across desktops, tablets, and smartphones. The visual design incorporates a vibrant Colour Hunt palette and a Pinterest background image, creating a fun, education-inspired aesthetic.

Objectives

- **Enable Quiz Selection** Allow users to choose from various topics and start quizzes with randomized questions.
- **Support Interactive Answering** Provide multiple-choice options, instant feedback on answers, and score tracking.
- **Create a Progress Tracker** Display real-time progress, timers, and final score summaries.
- **Demonstrate Front-End Development Skills** Showcase integration of HTML5 forms, JavaScript logic, and responsive design.
- **Ensure Cross-Device Compatibility** Provide seamless performance on laptops, desktops, tablets, and mobile devices.

Scope of the Project

Defining the scope of the project, the Interactive Quiz App serves as a lightweight, browser-based educational tool for personal or demo use. It demonstrates how modern browser APIs (local Storage for saving progress) can deliver interactive quizzes without external plugins. The scope includes question display with options, real-time feedback and scoring, a timer for each quiz, a responsive layout with modal results, and a consistent design using a Colour Hunt theme (FF6D60, F7D060, F3E99F, 98D8AA) and a Pinterest-sourced background. Future extensions may include user authentication, multiplayer modes, or integration with external question databases.

Tools & Technologies Used

- HTML5 Structure of the page, <form> elements for quiz questions and inputs.
- CSS3 & Bootstrap 5 Styling, responsive grid system, and modal dialogs.
- JavaScript (ES6) Quiz logic for randomization, scoring, and timer via set Interval ().
- jQuery Simplified DOM manipulation and event handling.
- Colour Hunt Palette & Pinterest Image Aesthetic theme and background.
- Visual Studio Code editing and project management.

System Design & Structure

- Quiz Section: Contains dynamic <div> elements for displaying questions, options, and timers.
- Control Panel: Buttons to start quizzes, submit answers, and navigate to next questions.
- Progress Bar: Responsive Bootstrap progress component to show quiz advancement.
- Modal Summary: Bootstrap modal displays final scores and feedback on completion.

CSS Styling Strategy

- Colour Palette: Vibrant coral (FF6D60) background with Pinterest texture overlay. Accents: F7D060 and F3E99F for buttons and borders. Highlights: 98D8AA for text and correct answer effects.
- Layout: Bootstrap Grid + Flexbox for responsive adaptation.
- Hover Effects: Smooth colour changes and shadow transitions on answer options.
- Modal Design: Rounded corners and light-theme overlay for clear result previews.

Key Features

- Quiz Topic Selection Users choose from predefined topics with randomized questions.
- Instant Feedback Highlights correct/incorrect answers immediately after submission.
- Timer & Scoring Counts down time per question and tracks overall score.
- Progress Tracking Visual bar shows quiz completion percentage.
- Modern UI Pinterest background and Colour Hunt theme for an engaging, education friendly appearance.

Challenges & Solutions

- Randomizing Questions Used JavaScript arrays and Math.random() to shuffle questions without repetition.
- Implementing Timer Utilized setInterval() for countdowns and clearInterval() on submission.
- Maintaining Responsiveness Combined Bootstrap grid and custom CSS to adapt to all screen sizes.
- Saving Progress Used local Storage to persist scores and quiz state without a server.

Outcome

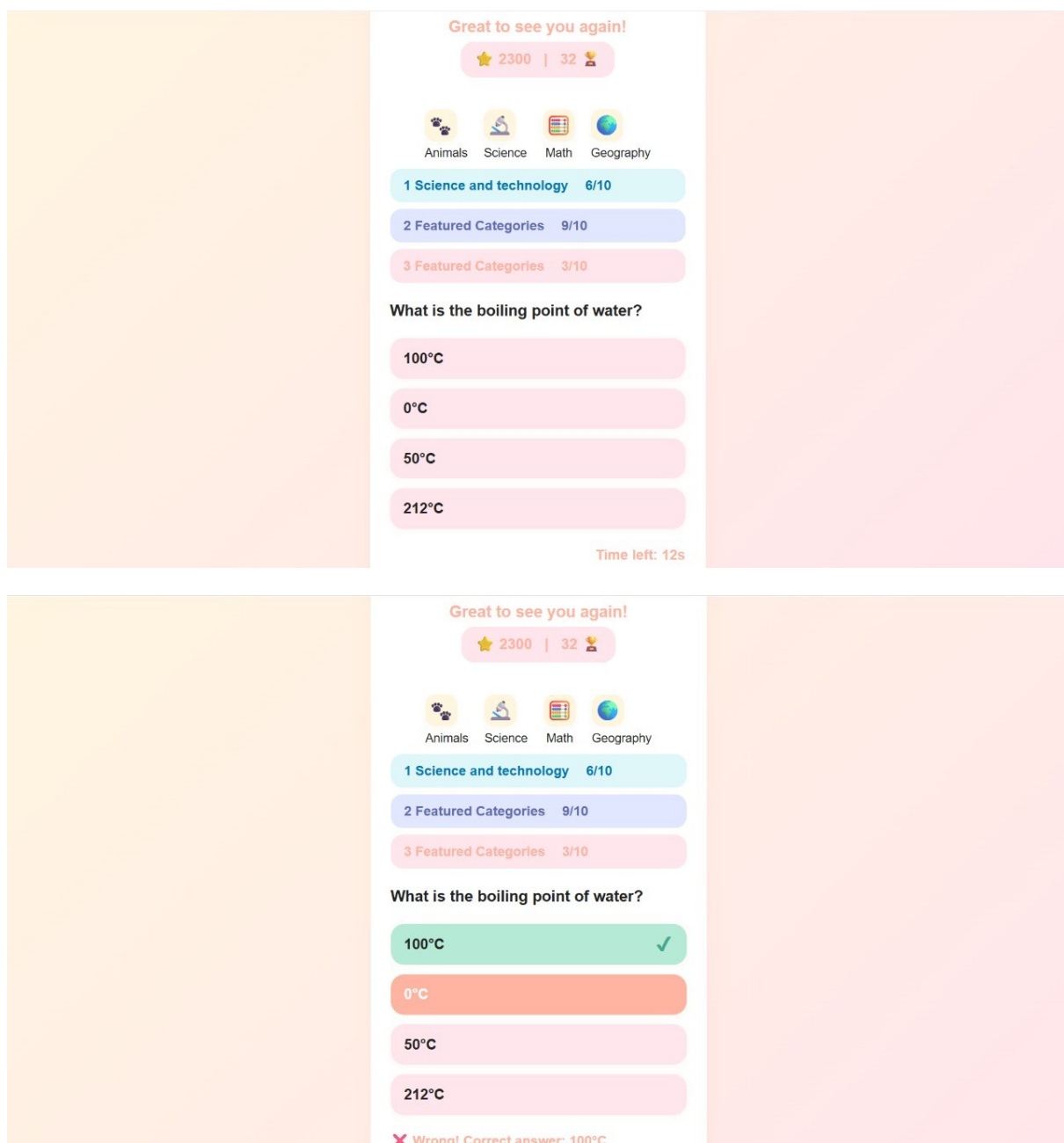
Describing the outcome, a functional quiz web app that loads questions, provides interactive answering, tracks progress, and displays scores. Smooth user experience with instant feedback

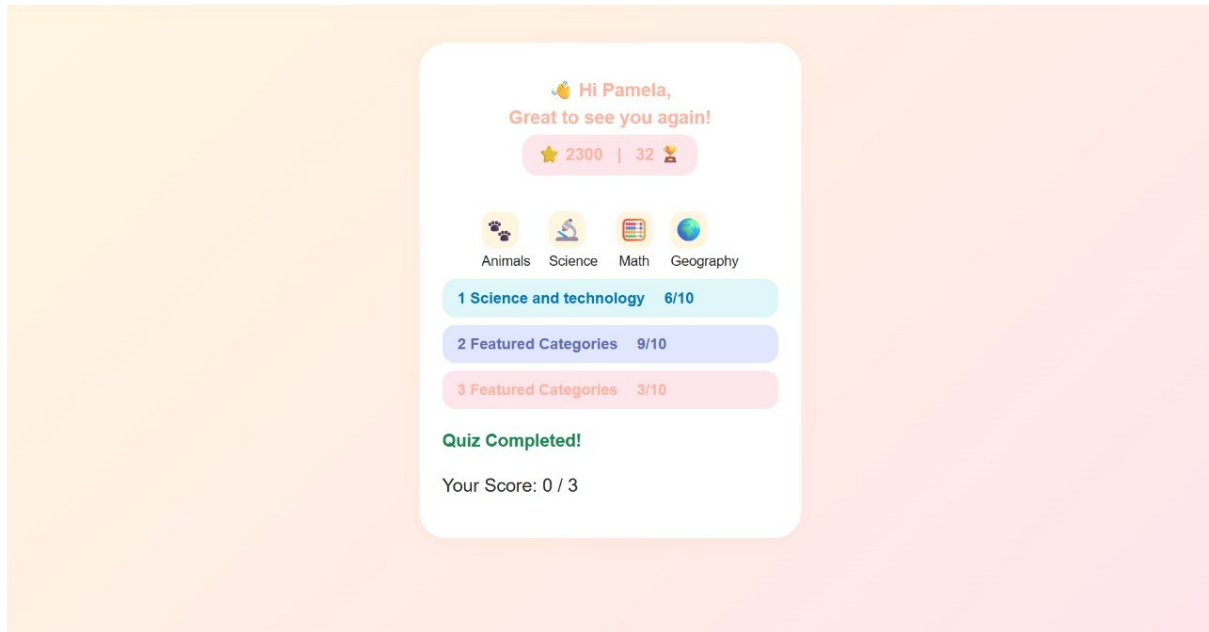
and modal summaries for completed quizzes. A visually engaging interface leveraging a modern color scheme and background texture.

Future Enhancements

- Add question types (true/false, fill-in-the-blank).
- Implement server-side storage to save high scores.
- Integrate adaptive difficulty using AI libraries like TensorFlow.js.
- Provide multi-language support for broader accessibility.

ScreenShots





Conclusion

Concluding the project, the Interactive Quiz App project demonstrates how HTML5, CSS3, Bootstrap, and JavaScript can be combined to create a feature-rich, interactive educational web application without additional plugins. It serves as a practical example of leveraging modern browser APIs to provide real-time user interaction in a responsive, aesthetically pleasing environment.